

LLMSpoofGuard: Real-Time Detection and Situational Awareness of GPS Spoofing in Aviation via ADS-B Data

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Abstract—Automatic Dependent Surveillance–Broadcast (ADS-B) is a critical component of modern aviation surveillance, enabling real-time tracking of civil aircraft worldwide. However, its reliance on satellite navigation systems makes it vulnerable to GPS spoofing, which can corrupt positional information and pose serious operational and safety risks. Existing GPS spoofing detection approaches largely rely on rule-based heuristics or supervised models, limiting their ability to generalize to previously unseen or evolving spoofing behaviors.

We present LLMSpoofGuard, a deployed GPS spoofing detection and situational awareness framework for civil aviation based on ADS-B data. The system is designed for operational monitoring, combining LLM-based trajectory reasoning, geospatial aggregation, and analyst-facing exploration tools. LLMSpoofGuard has been deployed in live aviation security workflows since November 2025, processing millions of ADS-B messages daily. The pipeline consists of four stages: (1) ingestion and preprocessing of live ADS-B streams enriched with aircraft metadata; (2) live-stream identification of physically implausible trajectory behavior via few-shot, constraint-aware LLM reasoning; (3) aggregation of detected events into dynamic geographic spoofing zones; and (4) interactive analysis through analyst-facing geospatial and agentic chat-based interfaces.

Unlike static detection models, the proposed framework emphasizes adaptability, interpretability, and human-in-the-loop validation, enabling analysts to monitor evolving spoofing behavior without frequent model retraining, leveraging LLM-driven classification via prompt-based reasoning. Offline evaluation on a real-world dataset of over 60,000 flight trajectories across 100+ countries demonstrates high agreement with a conservative known-pattern benchmark while also surfacing heuristic-bypassing anomalies that fixed rules miss. Furthermore, several apparent false positives under the rule-derived proxy labels were later reviewed by aviation security analysts and found to represent operationally suspicious, previously uncharacterized spoofing behaviors. Post-deployment observations highlight the system’s suitability for high-throughput, safety-critical environments.

Index Terms—GPS Spoofing, ADS-B Security, LLM-based Anomaly Detection, Clustering, and Aviation Security

I. INTRODUCTION

Global civil aviation increasingly relies on satellite-based navigation and broadcast surveillance technologies, making Global Positioning System (GPS) spoofing a growing operational risk [1]. In recent years, spoofing incidents have been observed across multiple regions, disrupting aircraft navigation and degrading situational awareness for air traffic controllers

and aviation security stakeholders [1]–[3]. Unlike isolated sensor faults, GPS spoofing often manifests as coordinated, region-wide effects impacting multiple aircraft simultaneously, requiring analysts to consider spatial, temporal, and semantic patterns rather than individual trajectory anomalies.

Prior research on Automatic Dependent Surveillance–Broadcast (ADS-B) [4] anomaly and spoofing detection has primarily focused on offline analysis and per-flight classification, using rule-based heuristics, supervised learning, or sequence models trained on historical data [5]–[9]. While these approaches can achieve strong detection accuracy under controlled conditions, deploying them in real-world aviation environments poses significant challenges. Spoofing tactics evolve, labeled data is limited and region-specific, and purely numerical anomaly scores provide limited support for human analysts who must interpret, validate, and respond to detected events within tight operational constraints.

These challenges motivate system designs that emphasize adaptability, interpretability, and human-in-the-loop analysis over purely automated detection [10].

This paper presents LLMSpoofGuard, a deployed GPS spoofing detection and situational awareness framework used continuously by professional analysts since late 2025. The framework ingests live ADS-B streams, builds 90-minute trajectory windows, detects spoofing behavior without retraining, aggregates events into geographic spoofing zones, and exposes the results through analyst-facing interfaces for exploration, comparison, and investigation (see Figure 1).

Rather than treating spoofing detection as an isolated classification task, LLMSpoofGuard emphasizes operational usability: detection outputs are aggregated into zone-level representations aligned with how analysts examine airspace risk, exposed through an interactive geospatial interface and a chat-based system for natural-language exploration of live and historical activity across regions, aircraft families, and temporal trends. A key operational challenge is identifying previously uncharacterized anomalous behaviors consistent with spoofing-like patterns without frequent model retraining or manual rule updates. To address this challenge, LLM-SpoofGuard employs LLM-based classification to identify potentially novel spoofing effects. Importantly, novelty signals are presented as anomaly notifications, enabling conservative

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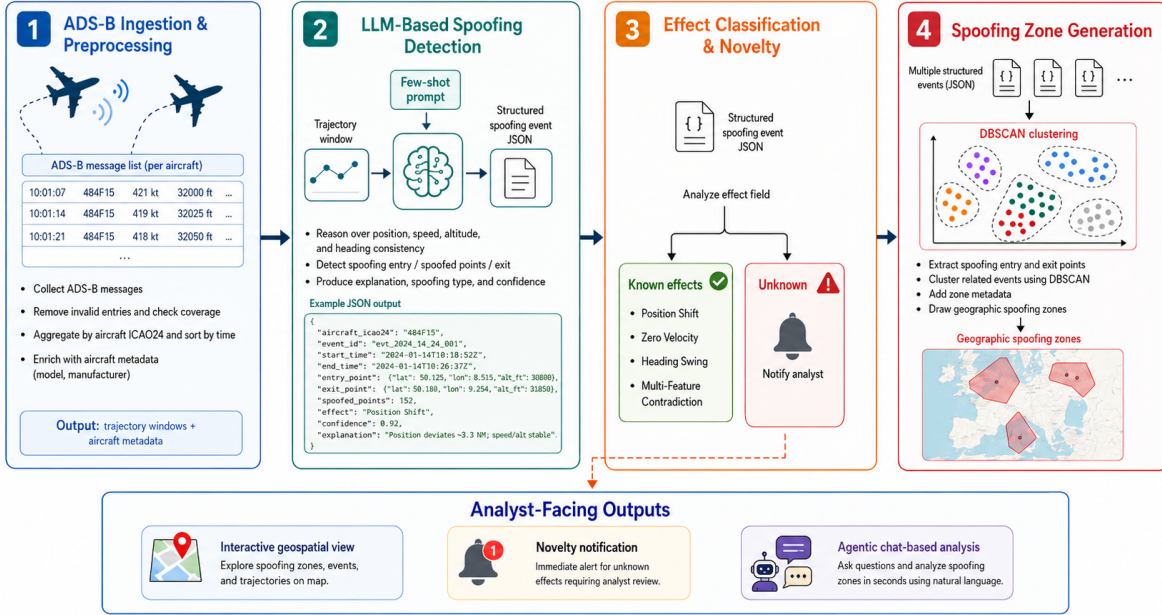


Fig. 1. LLMSpoofGuard architecture from ADS-B collection and preprocessing through LLM detection, spoofing-effect classification, zone generation, and analyst-facing outputs.

human-in-the-loop investigation in safety-critical settings.

To evaluate the effectiveness of LLMSpoofGuard, we leverage a comprehensive, real-world ADS-B dataset consisting of over 60,000 proxy-labeled flight trajectories across more than 100 countries. This global coverage allows us to evaluate the framework under diverse regional conditions. Our offline evaluation shows high agreement with conservative known-pattern proxy labels, strong geospatial zone quality, and the ability to flag previously uncharacterized anomalous behaviors. Furthermore, we provide post-deployment insights from sustained operational use, including system throughput metrics and patterns of analyst interaction.

This paper makes the following **contributions**:

- A deployed GPS spoofing monitoring system operating continuously in a safety-critical aviation security workflow since November 2025, ingesting approximately 6.48 million ADS-B states per day and analyzing about 9,000 trajectories per 90-minute window.
- An end-to-end operational pipeline that integrates trajectory-level LLM reasoning, density-based geospatial aggregation, and analyst-facing visualization and analytics. The contribution lies not in any single component (each draws from established techniques) but in their composition into a system that meets the operational constraints of throughput, interpretability, and analyst workflow integration in a domain with limited ground truth.
- An agentic chat-based analytical interface in which a central orchestrator selects live retrieval, historical database querying, or analytical reasoning tools and returns a

single contextual response, reducing time-to-insight from tens of minutes to seconds.

- A novelty identification mechanism that explicitly surfaces uncharacterized spoofing patterns for analyst review rather than forcing classification into known categories. We show that several LLMSpoofGuard detections counted as false positives under conservative rule-derived labels were later validated by analysts as spoofing-relevant, demonstrating discovery beyond fixed heuristics and known patterns.
- Comprehensive evaluation evidence, combining offline benchmarking on a globally distributed dataset of over 60,000 proxy-labeled trajectories with post-deployment analyst validation and spoofing zone comparison over a 60-day operational period. We release the deployed prompt, labeled trajectory identifiers and labels, and reference baseline implementations to support reproducibility within the constraints of operational deployment.

This work positions spoofing detection as a joint reasoning and aggregation problem over spatio-temporal data streams, extending beyond traditional point-wise anomaly detection paradigms.

II. BACKGROUND

Automatic Dependent Surveillance–Broadcast (ADS-B) is a core aircraft surveillance mechanism in which aircraft broadcast identity, position, altitude, velocity, heading, and timestamp information, largely derived from GNSS sources such as GPS [4], [11], [12]. While ADS-B improves situational awareness and air traffic management, its reliance on

satellite navigation exposes it to external interference [11], [13], [14]. In GPS jamming, attackers disrupt GNSS reception, often causing missing or invalid position reports [15]–[17]. In contrast, GPS spoofing injects counterfeit signals that produce syntactically valid but semantically false aircraft states, creating risks such as misrouting and degraded situational awareness [1], [15], [18], [19]. Operational spoofing detection presents three challenges. First, spoofing tactics are non-stationary and vary across regions, creating a distribution shift for static models. Second, authoritative ground truth is scarce, limiting supervised training and objective cross-regional evaluation. Third, analysts require interpretable, region-level evidence rather than opaque per-trajectory anomaly scores. LLM-SpoofGuard is designed around these constraints by combining anomalous flight-behavior reasoning with zone-level aggregation to provide actionable intelligence for aviation security.

III. RELATED WORK

LLMSpoofGuard intersects three research areas: ADS-B security, trajectory anomaly detection, and LLM-based operational analytics. Early ADS-B security methods focused on physical-layer verification using signal-level attributes such as Time-of-Arrival and Doppler shifts [20]–[23]. More recent work analyzes ADS-B trajectories directly, using rule-based flight-dynamics checks [5], LSTM encoder–decoders [6], [24], GRUs [25], Transformers [8], and VAE-based anomaly models [7]. These approaches can detect trajectory anomalies but typically operate on isolated flights and provide limited support for identifying coordinated, region-wide spoofing activity.

Geospatial aggregation methods address this gap by grouping individual anomalies into spatial units. Density-based clustering, especially DBSCAN [26], has been used for flight anomaly and trajectory analysis [27]–[30]. LLM-SpoofGuard uses this idea operationally, converting trajectory-level detections into dynamic spoofing zones that support live situational awareness. Finally, recent studies show that LLMs can support anomaly detection and reasoning over structured logs and time-series data [31]–[34], including zero-shot and few-shot settings [35], [36]. Unlike standalone classifiers, LLM-SpoofGuard uses an LLM as a physically constrained reasoning layer that synthesizes multiple flight-behavior signals into structured, analyst-interpretable spoofing outputs and supports discovery of heuristic-bypassing effects.

IV. SYSTEM ARCHITECTURE AND OPERATIONAL WORKFLOW

A. Problem Statement

We formulate GPS spoofing detection in civil aviation as a multi-level data mining problem that extends beyond traditional point-wise anomaly detection. Existing approaches primarily operate at the level of local kinematic deviations (e.g., velocity or altitude changes), capturing *syntactic anomalies* but failing to detect attacks that preserve message validity while violating physical feasibility over time.

In practice, sophisticated spoofing produces trajectories that are syntactically correct but *semantically inconsistent* with

aircraft dynamics. Furthermore, spoofing is often a regional phenomenon affecting multiple aircraft simultaneously, introducing a *unit-of-analysis mismatch*: prior methods analyze individual trajectories independently, ignoring spatio-temporal correlations across flights.

To address these limitations, a robust detection framework must jointly solve two tasks: (1) **Trajectory-Level Reasoning**: identifying semantically implausible behavior by integrating multi-dimensional flight state sequences; and (2) **Geospatial Aggregation**: consolidating trajectory-level detections into coherent spatial-temporal regions (“spoofing zones”) that reflect coordinated interference and support operational decision-making. This formulation defines spoofing detection as a joint reasoning and aggregation problem over structured spatio-temporal data streams.

B. End-to-End System Overview

LLMSpoofGuard is deployed as a live aviation security monitoring system that continuously ingests ADS-B streams, detects GPS spoofing behavior, aggregates detected events into geographic spoofing zones, and exposes the results through analyst-facing interfaces. The system operates on consecutive 90-minute analysis windows: ADS-B states are sampled every two minutes, yielding up to 45 states per aircraft, and approximately 9,000 valid trajectories are analyzed at the end of each window.

The system’s end-to-end operational pipeline is presented in Figure 1. Incoming live ADS-B messages are collected using the OpenSky Network (OSN) public API, preprocessed, and grouped into 90-minute trajectory windows. These trajectories are then analyzed by an LLM-based spoofing detection module, which outputs structured spoofing indicators and candidate spoofing points. Detected spoofing events are aggregated spatially and temporally into spoofing zones, which represent areas of high-confidence operational risk. All detected zones and associated metadata are stored in a database and consumed by multiple analyst-facing applications, including an interactive geospatial view and a chat-based analytical interface.

C. Preprocessing

ADS-B messages are collected from live surveillance feeds at a fixed two-minute sampling interval and transformed into 90-minute (up to 45 messages) trajectory-level representations used in both offline evaluation and live deployment. Each trajectory is modeled as a time-ordered sequence of state vectors comprising the geographic position (latitude, longitude), barometric altitude (m), geometric altitude (m), ground speed (m/s), heading, and Unix timestamp. Barometric altitude is included, as it remains unaffected by GNSS spoofing and provides a stable reference for vertical consistency checks.

Raw ADS-B streams are segmented into continuous trajectories, resampled to a consistent two-minute resolution, and filtered to remove low-quality sequences caused by receiver dropouts or coverage gaps. Trajectories with fewer than 10 consecutive timestamps are discarded to avoid unreliable short

windows. Each trajectory is enriched with aircraft manufacturer and model information, which is retrieved from external aircraft registries using ICAO24 identifiers.

This preprocessing pipeline is applied identically in both offline evaluation and live operation, ensuring consistency between experimental results and deployed system behavior.

D. LLM-Based Spoofing Detection

The detection module analyzes ADS-B trajectory segments and determines whether they exhibit GPS spoofing behavior. Each trajectory is represented as a time-ordered sequence of state vectors, as described in the preprocessing phase above.

Rather than relying on pretrained sequence models or hand-crafted rules, LLMSpoofGuard employs a few-shot [36] LLM, which is responsible for reasoning about physically implausible flight behavior. The LLM receives a structured prompt containing trajectory summaries and contextual features and produces a structured output containing the spoofing decision, entry/exit points, spoofed segment, effect category, confidence, and a natural-language explanation.

The model is explicitly constrained to reason over physically grounded features (e.g., velocity, altitude, and heading consistency), ensuring that predictions remain tied to domain-valid flight dynamics rather than purely statistical patterns.

1) *Prompt Design:* To bridge the gap between numerical ADS-B streams and the LLM’s reasoning capabilities, LLMSpoofGuard employs a structured trajectory serialization process. Raw, high-frequency ADS-B message logs are not passed directly to the model. Instead, each 90-minute trajectory window is summarized as a compact sequence of state transitions. LLMSpoofGuard uses a fixed, structured prompt that constrains the LLM to assess physical feasibility in aircraft trajectories rather than perform anomaly detection. The prompt specifies a single aircraft trajectory as a sequence of consecutive ADS-B state transitions, augmented with aircraft manufacturer and model metadata to establish performance constraints. Although reasoning is constrained to local transitions, the model implicitly captures temporal inconsistencies through aggregated multi-step reasoning across consecutive segments. The model is instructed to evaluate violations of physical consistency across position, velocity, heading, and cross-feature relationships, while accounting for rare but legitimate operational maneuvers through fixed counterexamples. When spoofing is detected, the model delineates the event boundaries and assigns a categorical effect label from a closed set, or returns a label of “Unknown” to represent uncertainty.

E. Spoofing Zone Generation

Detected spoofing events are aggregated into geographic spoofing zones, which form the primary unit of analysis for operational monitoring. A spoofing zone represents a spatial region and time interval in which multiple spoofing events—possibly affecting multiple aircraft—are observed.

For each spoofing event detected, the system extracts the spoofing entry and exit points and applies a density-based clustering algorithm to group spatially and temporally related

events. We employ DBSCAN [26] due to its ability to identify non-convex clusters and operate without the need to specify the number of clusters in advance. Clustering is performed on geographic coordinates, with temporal constraints to prevent merging unrelated events across distant time windows.

Zone-level aggregation provides several operational advantages. First, it reduces alert fatigue by consolidating multiple trajectory-level detections into a single regional signal. Second, it enables analysts to reason about spoofing activity at the level of airspace regions rather than individual flights. Third, it supports longitudinal analysis of zone persistence, expansion, and migration over time. The zone generation process was designed to be independent of the underlying detection model, agnostic to how the detections are produced.

F. Identification of Novel Spoofing Effects

For each spoofing-event output described in Section IV-D, the LLM assigns one or more known spoofing-effect categories (e.g., position shift, velocity drop, altitude drop), or returns “Unknown” when no known category fits. The category bank is provided in the system prompt and reflects spoofing patterns from the literature. An Unknown designation is treated as an explicit uncertainty signal for analyst review rather than an error.

G. Analyst-Facing Applications

While spoofing detection and zone generation operate automatically, LLMSpoofGuard exposes detected zones through a geospatial map, statistical overview, and agentic chat interface. The map supports live and historical filtering by time, aircraft type, and effect category. In the chat workflow, a central orchestrator interprets each query in conversational context and selects one of four actions: live zone retrieval, historical database querying, analytical reasoning over retrieved data, or a direct response without tool use. Tool outputs return to the orchestrator context, and only the orchestrator produces the analyst-facing response. This complements the map for cross-region, historical, and multi-variable questions that would otherwise require several manual filters or queries.

V. DATASET AND ANNOTATION

A. Data Collection

We evaluate LLMSpoofGuard using a large-scale, real-world dataset of civil aviation trajectories sourced via the OpenSky Network [37]. The dataset comprises approximately 1.15 million ADS-B messages from 61,565 trajectory segments of varying lengths (10-45 ADS-B messages), covering over 70 aircraft models across 100+ countries. Data was sampled across multiple temporal windows between November 2024 and June 2025 to capture diverse operational conditions, noise profiles, and regional variability. Figure 2 illustrates a representative spoofing event, highlighting the divergence between reported and intended flight paths.

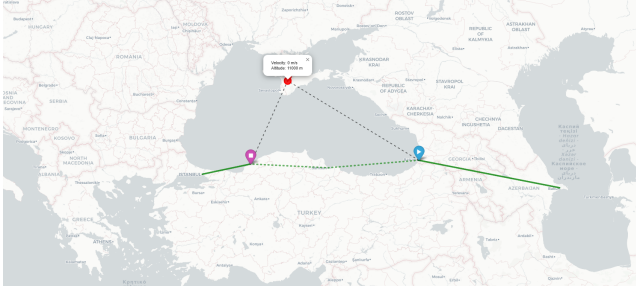


Fig. 2. GPS spoofing effect on a Baku-to-Istanbul flight. Solid green lines: reported trajectory segments. Dashed green line: filed flight plan (intended path). Red marker: estimated spoofing onset. Black dashed lines: physically implausible positional jumps.

TABLE I
RULE-BASED HEURISTICS (RBH) FOR CONSERVATIVE KNOWN-PATTERN PROXY LABELS.

Rule	Condition
Altitude Drop	Decrease > 4000 m within 2 min.
Timestamp Freeze	Same timestamp across ≥ 3 consecutive updates.
Zero Velocity	Ground speed < 50 m/s while altitude > 5000 m.
Heading Change	$ \Delta\text{heading} > 120^\circ$ between consecutive updates.
Position Shift	$> 1.8^\circ$ lat/lon or > 250 km great-circle within 2 min.
Altitude Increase	Increase > 3000 m within 2 min outside takeoff.

B. Annotation: Conservative Known-Pattern Labels

Establishing definitive ground truth for real-world GPS spoofing is inherently challenging because authoritative incident logs are rarely available on a global scale. We therefore construct a conservative proxy-label benchmark designed to capture high-confidence, physically grounded spoofing patterns while explicitly preserving a separate channel for heuristic-bypassing behavior.

a) *Tier 1: Known-pattern rule labels:* We define six rule-based heuristics (RBH) (see Table I) that identify explicit kinematic violations associated with known spoofing effects. These rules are based on domain knowledge and are used to construct the conservative known-pattern proxy labels for offline benchmarking. RBH is therefore treated as an oracle/reference for this proxy label set, not as an independent competing detector. All pair-wise rules are evaluated only when the time gap between two consecutive ADS-B updates satisfies $\Delta t \leq 120s$. Absolute-threshold rules that do not require a delta (zero velocity, timestamp freeze) are additionally evaluated up to $\Delta t \leq 150s$. Larger gaps are treated as reasoning boundaries and skipped, because resampled gaps above 2.5 minutes can introduce artificially large kinematic deltas that are not attributable to spoofing.

A trajectory is assigned its dominant rule label as the most frequent point-level category triggered along it; trajectories where no rule fires are labeled clean.

Spoofing behavior that violates physical feasibility while remaining below every individual threshold is, by construction,

not labeled positive by RBH; such cases are surfaced by the LLM via the Unknown label (Tier 2).

b) *Tier 2: Heuristic-bypassing discovery labels:* The RBH rules intentionally cover only known, explicit threshold violations. However, spoofing behavior may remain below individual rule thresholds while still producing physically inconsistent trajectories over multiple consecutive segments. LLMSpoofGuard is designed to surface these cases through an *Unknown* label rather than forcing them into the predefined category bank. A representative example identified during evaluation and deployment is *unrealistic velocity behavior*, where reported ground speed exceeds physically plausible civil aviation limits while other trajectory channels remain locally consistent. Such events are initially outside the RBH proxy label set and may therefore appear as false positives in offline evaluation, but are subsequently reviewed by aviation security analysts. This labeling design is intentionally conservative. The offline benchmark measures agreement with known-pattern proxy labels rather than absolute spoofing ground truth. Consequently, LLMSpoofGuard detections outside the RBH rule set are not automatically interpreted as errors; they are treated as candidates for analyst validation and potential expansion of the spoofing-effect taxonomy.

C. Geographic Distribution

The dataset spans 100+ countries and includes both high-intensity spoofing regions, such as the Black Sea, Eastern Mediterranean, and Baltic Sea, and lower-risk airspace, such as India, Australia, and the South China Sea. Observed spoofing prevalence varies substantially across regions, ranging from 87.5% in the Eastern Mediterranean to less than 1% in North America and Australia. This diversity reduces the likelihood that evaluation results are driven by a single region-specific spoofing pattern.

D. Operational Consistency

The same preprocessing and feature representation are used in offline evaluation and live deployment, minimizing the sim-to-real gap. Because authoritative global spoofing ground truth is unavailable, the offline benchmark measures agreement with conservative proxy labels and is complemented by post-deployment analyst validation (Section VII).

VI. OFFLINE EVALUATION AND VALIDATION

Offline evaluation was conducted to assess two complementary properties of LLMSpoofGuard before and during

TABLE II
DISTRIBUTION OF SPOOFING EFFECT CATEGORIES IN THE PROXY-LABELED DATASET.

Effect Category	Prevalence (%)
Heading change	24.9
Zero velocity	13.6
Altitude increase	13.1
Position shift	12.5
Unknown	12.5
Timestamp freeze	11.9
Altitude drop	11.5

deployment: (i) agreement with conservative known-pattern spoofing labels, and (ii) the ability to surface heuristic-bypassing behaviors that are not captured by fixed rules. All experiments use the dataset described in Section V and the same preprocessing pipeline used in deployment.

A. Baselines and Evaluation Protocol

We compare LLMSpoofGuard against representative ADS-B anomaly-detection baselines, including LSTM reconstruction [6], [38], Isolation Forest [39], XGBoost [40], and the rule-based heuristic procedure (RBH) defined in Section V-B. All learning-based baselines are trained and evaluated over five random seeds (42–46), each with an independent 80/20 stratified split; we report mean $\pm$ standard deviation across seeds. GPT-4.1 mini is evaluated once at temperature 0, with Wilson 95% CIs over its per-trajectory predictions. The LSTM baseline is a two-layer encoder–decoder trained to reconstruct sequences over full 45-message trajectories, following the architecture of Habler et al. [6]. XGBoost-point and Isolation Forest operate on local kinematic deltas between consecutive ADS-B messages. XGBoost-Traj is a stronger trajectory-level supervised baseline that summarizes each 90-minute window using global trajectory statistics, transition-level delta statistics, physics-consistency indicators, and aircraft metadata. Importantly, RBH is not treated as a competing detector in the usual sense. All baseline implementations, model initialization, hyperparameter search ranges, selection criteria, and final configurations are provided in the released repository. The RBH rules define the conservative known-pattern proxy label set used for offline benchmarking. Therefore, RBH obtains perfect agreement by construction and is reported only as an oracle/reference for fixed-threshold detection on known spoofing categories. This proxy-label design is intentionally conservative: spoofing behaviors outside the RBH rule set are initially counted as negatives, even if they may correspond to genuine but previously uncharacterized spoofing effects.

B. Known-Pattern Detection Performance

Table III reports performance on the conservative known-pattern proxy benchmark. Among non-oracle methods, LLM-SpoofGuard achieves the strongest agreement with the proxy labels, reaching 98% proxy-label accuracy and 97% F1. This result indicates that the LLM-based detector captures the known physical violation patterns represented by RBH while also producing structured spoofing metadata for interpretability and downstream zone aggregation.

The main advantage of LLMSpoofGuard is not only its high agreement with known rule-derived labels, but also its behavior in disagreement cases. Because the offline labels are derived from fixed known-pattern rules, LLM-positive/RBH-negative cases are counted as apparent false positives. However, the analyst review showed that several of these disagreements correspond to spoofing-relevant behaviors outside the original rule set, including velocity spikes and coordinated multi-channel drift. The 98% proxy-label accuracy should therefore be interpreted as a conservative estimate of

TABLE III
DETECTION PERFORMANCE ON THE CONSERVATIVE KNOWN-PATTERN PROXY BENCHMARK. RBH DEFINES THE PROXY LABELS AND IS THEREFORE SHOWN AS AN ORACLE/REFERENCE RATHER THAN AS A COMPETING DETECTOR.

Model	Accuracy (%)	Precision (%)	Recall (%)	F1 (%)
LSTM (S; T)	79.4	75.0	80.3	77.6
XGBoost-point (S; P)	64.3	55.5	96.9	71.3
XGBoost-Traj (S; T)	85.4	80.4	88.6	84.3
Isolation Forest (U; P)	72.2	62.2	94.3	74.9
LLMSpoofGuard (GPT-4.1 mini, F)	98.0	95.0	99.0	97.0
<i>Oracle/reference: RBH obtains 100% by construction.</i>				

agreement with known spoofing patterns, while the remaining disagreement channel provides evidence of discovery beyond fixed heuristics. Examining the false positives of the other baselines reveals a qualitatively different pattern. The false positives of XGBoost-Traj and Isolation Forest are dominated by legitimate trajectories — primarily takeoff and landing segments whose large kinematic deltas (rapid altitude change, heading reorientation, low ground speed) cause these models to misclassify them as anomalous. In other words, the LLM’s residual disagreement with RBH concentrates in the heuristic-bypassing channel that motivates the Tier-2 design, whereas the supervised and unsupervised baselines’ residual errors concentrate on phase-of-flight artifacts that carry no operational signal.

This pattern may reflect a difference in how the models use trajectory context. Supervised baselines fit point-wise or aggregated kinematic features against training samples, making them sensitive to label distribution and prone to confusing legitimate phase-of-flight dynamics with anomalies. In contrast, the LLM reasons over the entire trajectory, identifies the operational flight phase (takeoff, cruise, descent), and evaluates point-wise dynamics against both physical constraints and few-shot exemplars. This phase-aware, rule-and-example-grounded reasoning is not represented directly by the fixed-feature baselines and may contribute to the higher RBH agreement and different error patterns.

Beyond raw detection accuracy, the LLM-based approach also directly outputs structured spoofing metadata, which is essential for interpretability and downstream spoofing zone aggregation.

C. LLM Model Selection and Prompt Design

We evaluate both prompt design and model choice to quantify the trade-off between detection accuracy, latency, and cost. First, we compare the deployed few-shot prompt with a zero-shot variant using the same trajectory representation and output schema. As shown in Table IV, few-shot prompting improves recall from 65.0% to 99.0% and F1 from 75.0% to 97.0%. This suggests that examples are important for calibrating physical-consistency reasoning and reducing missed detections under the conservative proxy benchmark. We then evaluate several LLM variants using the deployed few-shot prompt. Table V reports accuracy, average inference time, and token cost. Open-weight models were evaluated locally on a single A100 GPU,

TABLE IV
PROMPT CONFIGURATION ABLATION FOR LLMSPOOFGUARD.

Prompt setting	Acc.	Prec.	Rec.	F1
Zero-shot	82.5	85.0	65.0	75.0
Few-shot	98.0	95.0	99.0	97.0

TABLE V
PERFORMANCE, INFERENCE TIME, AND TOKEN COST OF LLM VARIANTS FOR TRAJECTORY SPOOFING DETECTION. MODEL VERSIONS AND PRICING REFLECT THE APIS AVAILABLE AT THE TIME OF DEPLOYMENT AND ARE INTENDED AS A DEPLOYMENT-TIME COMPARISON RATHER THAN A STABLE BENCHMARK. OPEN-WEIGHT MODELS WERE EVALUATED LOCALLY ON A SINGLE A100 GPU. INFERENCE TIME EXCLUDES NETWORK LATENCY FOR CLOUD-HOSTED MODELS.

LLM	Accuracy (%)	Avg. inference time (s)	Input cost per 1M tokens (\$)	Output cost per 1M tokens (\$)
GPT-4.1	98	0.70	2.00	8.00
GPT-4o	97	1.50	2.50	10.00
GPT-4.1 mini	98	0.49	0.40	1.60
o1-mini	99	2.60	1.10	4.40
GPT-5.2	99	2.20	1.75	14.00
Llama 3.1 8B	96	0.80	0.27	0.27
Mistral 7B	97	0.30	0.10	0.10
Qwen3-Max-Thinking	99	3.50	0.20	0.20

while GPT-based models were accessed through the OpenAI API. Based on this comparison, GPT-4.1 mini was selected for deployment. It matches GPT-4.1 proxy-label accuracy at 98%, has the lowest cloud-model latency in our evaluation, and reduces token cost substantially relative to larger GPT models. Although GPT-o1-mini, GPT-5.2, and Qwen3-Max-Thinking achieve slightly higher proxy-label accuracy, their added latency or cost provides limited operational benefit for continuous monitoring. GPT-4.1 mini therefore offers the best accuracy–cost–latency balance while preserving prompt-based adaptability: new categories, counterexamples, or analyst-confirmed Unknown patterns can be incorporated through prompt updates rather than model retraining.

D. Spoofing Zone Generation Evaluation

We evaluate zone generation by comparing DBSCAN with K-Means and by examining DBSCAN parameter sensitivity. DBSCAN is better suited to spoofing-zone construction because it captures irregular, non-convex geographic regions without requiring the number of zones in advance, while K-Means may split or merge distinct spoofing regions. Figure 3 shows the expected tradeoff: small ϵ values fragment zones, while large values produce overly broad regions. For deployment, we use $\epsilon = 250$ km and $\text{min_samples}=5$, selected from this sensitivity analysis.

E. Novel Spoofing Effects and RBH-Disagreement Analysis

The novelty mechanism is directly connected to the RBH-disagreement cases discussed above. Because RBH labels cover only predefined fixed-rule patterns, LLMSpoofGuard detections outside those rules appear as false positives in the offline benchmark. To determine whether these cases represent noise or meaningful operational signals, we manually reviewed all events assigned the Unknown label by LLMSpoofGuard.

Of the detected spoofing events in the offline evaluation, 12.5% ($n = 769$) were assigned the Unknown label. Three avi-

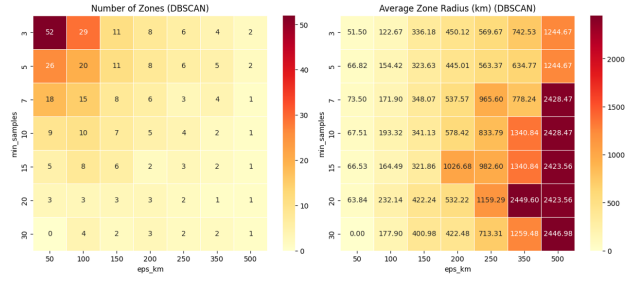


Fig. 3. Impact of DBSCAN parameters (ϵ , min_samples) on the number of detected zones and average zone radius.

TABLE VI
MANUAL VALIDATION OF UNKNOWN SPOOFING EFFECTS.

Outcome	Count	Share (%)
Outside known category bank	513	66.7
Variant of a known category, sub-threshold	184	23.9
Legitimate but unusual maneuver	51	6.6
Inconclusive, insufficient context	21	2.8

ation security analysts independently reviewed each Unknown-flagged trajectory and categorized the outcome, as shown in Table VI. Analysts reached unanimous agreement on 78% of cases, with the remainder resolved by majority vote (Fleiss' $\kappa = 0.80$), reflecting the inherent subjectivity of the four-way taxonomy. Among Unknown events, 66.7% were judged by analysts to be spoofing-relevant anomalies outside the known category bank, and an additional 23.9% were judged to be variants of known categories below RBH thresholds. Only 6.6% were judged legitimate but unusual maneuvers. LLMSpoofGuard therefore does not merely reproduce RBH decisions: it agrees with known spoofing patterns while surfacing heuristic-bypassing behaviors that fixed rules miss.

Table VII illustrates an analyst-confirmed unknown case where reported ground speed (800 m/s) exceeds the A380's maximum operating speed by $3\times$ while position-implied speed remains plausible — an anomaly invisible to all six RBH rules. The analyst-confirmed spoofing-relevant cases included three recurring patterns: *hyper-velocity inconsistency*, where reported ground speed exceeds civil airframe limits while position-implied speed remains plausible (as either single-sample spikes or sustained mismatches); *coordinated multi-channel drift*, where position, velocity, and heading co-drift over multiple segments while each local transition remains below fixed thresholds; and *heading-position desynchronization*, where heading changes rapidly while position remains nearly static. These cases demonstrate the operational value of the Unknown channel as a discovery mechanism and motivate future expansion of the spoofing-effect category bank.

VII. DEPLOYMENT AND POST-LAUNCH PERFORMANCE

A. Deployment Overview

LLMSpoofGuard has been continuously deployed using Amazon's Web Services since November 1, 2025. The system operates 24/7 and ingests live ADS-B data streams from

TABLE VII
EXAMPLE UNKNOWN CASE CONFIRMED BY ANALYSTS:
VELOCITY-CHANNEL SPIKE DURING SMOOTH CRUISE OF AN EMIRATES
AIRBUS A380 AT FL410.

Time	Lat.	Lon.	Alt. (m)	Rep. v (m/s)	Impl. v (m/s)	Hdg (°)
00:00:00	27.235	48.559	12,497	240.6	—	90.2
00:01:54	27.200	48.831	12,497	240.7	238.9	90.5
00:03:50	27.163	49.109	12,497	241.3	239.2	90.8
00:05:48	27.125	49.392	12,497	800.0	240.1	91.0
00:06:53	27.104	49.549	12,497	241.0	242.0	91.2

OpenSky Network, covering global civil aviation traffic. It is used by professional aviation security analysts at one of the world’s largest aircraft manufacturers as part of their routine monitoring for spoofing effects on their manufactured aircraft. The same zone-level outputs could also support ATC situational awareness, although integration with operational ATC workflows is left for future deployment. The framework has remained stable under sustained load since its deployment and has experienced no major service outages. To maintain monitoring continuity, the pipeline falls back to RBH-based detection during transient LLM API outages, ensuring uninterrupted analyst-facing zone updates. The agentic chat interface fails over to locally hosted LLMs.

B. Throughput, Scale, and Latency

In production, LLMSpoofGuard accumulates ADS-B states at two-minute intervals and analyzes the resulting trajectories once per 90-minute window. Each window contains approximately 9,000 valid trajectories, all of which are submitted to GPT-4.1 mini without a pre-filter in batches of 100. At approximately 10 seconds per batch, the roughly 90 batches account for about 15 minutes of LLM batch time, within the 90-minute monitoring cadence. Across the stream, this corresponds to approximately 6.48 million ADS-B states ingested per day. Table VIII summarizes these production measurements. The chat-based analytical interface exhibits predictable latency that scales with the query complexity. For typical analytical queries involving a limited number of spoofing zones, response latency is approximately 5–10 seconds, while larger historical queries may require up to 30 seconds end-to-end.

C. Operational Impact and Analyst Efficiency

Since its deployment, LLMSpoofGuard has shifted analyst workflows from trajectory-level alerting to region-level situational awareness.

TABLE VIII
POST-DEPLOYMENT FRAMEWORK PERFORMANCE.

Metric	Value
Valid trajectories/window	~9,000
ADS-B sample interval	2 minutes
Analysis window	90 minutes
LLM batch size	100 trajectories
LLM batch time	~10 seconds
ADS-B states/day	~6.48M
System outages since deployment	None

TABLE IX
POST-DEPLOYMENT ANALYST VALIDATION ON A STRATIFIED SAMPLE
($N = 1,000$). WILSON 95% CIs REPORTED.

Sample	Outcome	Rate [95% CI]
Flagged	Spoofing-relevant	99.0 [97.7, 99.7]
Flagged	Disagreement	1.0 [0.3, 2.3]
Non-flagged	Clean	98.0 [96.4, 99.0]
Non-flagged	Disagreement	2.0 [1.0, 3.6]
Explan./category	Consistent	100.0

Alert Fatigue Reduction via Zone Aggregation. We compare trajectory-level detection counts to post-DBSCAN zone counts over matched daily windows. Traditional ADS-B monitoring pipelines generate hundreds of daily trajectory-level alerts, whereas LLMSpoofGuard’s zone aggregation reduces the analyst-facing workload to 15–25 active zones per day, an approximate 90–95% reduction in raw alert volume.

Investigation Latency (Time-to-Insight). Manual investigation time reflects analyst-reported workflow estimates for representative regional spoofing investigations, while chat-based time-to-insight is measured as end-to-end response latency for equivalent natural-language queries. Manual investigation of regional spoofing events using raw ADS-B logs previously required approximately 30–60 minutes per incident. Using LLMSpoofGuard, natural-language queries (e.g., “Compare spoofing zones around France”) return a synthesized summary of aircraft impact and spoofing types in 5–10 seconds, a reduction of over three orders of magnitude in time-to-insight.

D. Post-Deployment Analyst Validation

To assess real-world detection reliability under deployment, we conducted an analyst-validated review of LLMSpoofGuard outputs over a 60-day period (Jan–Mar 2026), sampling 10–20 trajectories per day across diverse regions and times.

Evaluation Protocol. We randomly sampled 500 flagged and 500 non-flagged trajectories ($N = 1,000$), stratified to balance positive and negative outputs. Three aviation analysts independently labeled each trajectory as spoofing-relevant or clean based on trajectory consistency, regional context, and domain expertise, and judged whether the generated explanation and spoofing category were consistent with the observed behavior. To avoid confirmation bias, analysts labeled the spoofing-relevant/clean decision blind to the LLM’s category and explanation; the explanation/category consistency judgment was made in a separate pass after the primary label was finalized. Analysts reached full agreement on 92.5% of trajectories (925/1,000), with the remaining 7.5% resolved by 2-vs.-1 majority vote (Fleiss’ $\kappa = 0.90$, almost perfect agreement).

Because the sample is stratified rather than exhaustive, Table IX reports analyst-confirmation rates rather than unconditional recall, which would require uniform sampling over the full live stream—impractical given high traffic volume and low base rates. The explanation/category consistency metric reflects analyst judgment on reviewed confirmed cases and

is not a formal measure of causal correctness. Within this interpretation, flagged events are overwhelmingly confirmed as spoofing-relevant, and sampled non-flagged trajectories are predominantly clean. Moreover, apparent trajectory-level false positives have limited operational impact, as isolated detections lack spatio-temporal coherence and do not form spoofing zones.

E. External Corroboration via Public Spoofing Maps.

Because authoritative ground-truth labels for aviation GPS spoofing are rarely available at scale, we further evaluated whether LLMSpoofGuard detections align with an independent public spoofing-monitoring source. Specifically, we cross-referenced high-confidence spoofing zones detected during the same 60-day period against GPSwise.aero, a publicly available aviation GNSS-interference platform that reports live GPS spoofing and jamming regions. A zone was counted as corroborated if its polygon intersected a GPSwise-reported region with IoU > 0.70 or centroid distance below 100 km, and the two reports overlapped within ± 90 minutes. Out of 300 high-confidence zones examined, 287 (95.6%) exhibited spatial and temporal overlap with GPSwise-reported spoofing regions. This result should not be interpreted as definitive ground truth, since public interference maps may have incomplete coverage and may rely on overlapping aviation data sources. However, the agreement provides external corroboration beyond the proxy-label benchmark and the internal analyst review, supporting the operational plausibility of the detected zones.

Taken together, our evaluation follows a layered evidence design. The offline benchmark measures agreement with conservative known-pattern proxy labels; the Unknown/RBH-disagreement review evaluates whether rule-bypassing detections carry operational signal; the post-deployment study measures analyst confirmation on a stratified live sample; and the GPSwise comparison provides external spatial-temporal corroboration at the zone level. We therefore do not claim definitive forensic attribution for every event. Instead, the evidence supports operational spoofing detection and situational awareness under the ground-truth constraints typical of global ADS-B monitoring.

F. Limitations

Despite its stable deployment and strong empirical performance, LLMSpoofGuard has several limitations. First, the framework relies only on ADS-B broadcasts and does not directly observe intent-level information such as ATC clearances, onboard navigation states, or complete flight-plan updates. As a result, rare but legitimate maneuvers, including emergency deviations or atypical routing in constrained airspace, may occasionally appear spoofing-like.

Second, evaluation is constrained by the absence of an authoritative global ground truth for GPS spoofing. Our offline results, therefore, measure agreement with conservative, operationally validated proxy labels rather than definitive forensic confirmation. In particular, because the proxy benchmark is based on known rule-derived patterns, some detections outside

the RBH rule set are counted as false positives even when they may represent genuine but previously uncharacterized spoofing behavior. We mitigate this through analyst review of Unknown and RBH-disagreement cases, but definitive validation remains difficult without independent onboard or air-traffic-control ground truth. Third, subtle long-horizon attacks, such as slow drift spoofing, remain challenging. These attacks may preserve local physical consistency while gradually shifting the trajectory over time. Addressing this limitation will require longer-horizon reasoning and integration with external context, such as live NOTAMs, weather overlays, digital flight clearances, and filed flight plans.

VIII. CONCLUSION

This paper presented LLMSpoofGuard, a deployed GPS spoofing detection and situational awareness framework for civil aviation based on live ADS-B data. The system has been operating continuously since November 2025, processing millions of messages per day and supporting analyst-driven monitoring of spoofing activity at a global scale. Beyond detection accuracy, LLMSpoofGuard demonstrates the importance of end-to-end system design in applied data mining settings. By combining LLM-based trajectory reasoning, geospatial aggregation into spoofing zones, and analyst-facing interfaces, the framework transforms low-level trajectory anomalies into actionable, region-level intelligence. Our results show that LLM-based reasoning can effectively integrate multiple weak physical signals into decisions, achieving strong agreement with known rule-derived spoofing patterns while outperforming traditional machine learning baselines on the conservative proxy-label benchmark and surfacing heuristic-bypassing behaviors missed by fixed rules. Future work will focus on slow-drift spoofing and on integrating independent context such as NOTAMs, weather, digital flight clearances, and filed flight plans.

IX. REPRODUCIBILITY

To support reproducibility within the constraints of operational deployment, we release the code and experimental material at https://github.com/yf2595/LLMSpoofGuard_ICDM_ADS_2026, containing: (i) the full deployed LLM prompt, including the structured output schema, the closed-set category bank, the few-shot examples, and the legitimate-maneuver counterexamples; (ii) the labeled trajectory dataset as OpenSky-fetchable ICAO24 identifiers, time windows, and spoofing labels, together with a re-fetch script; (iii) reference implementations and hyperparameter configurations for all baselines (LSTM, XGBoost-point, XGBoost-Traj, Isolation Forest, RBH); and (iv) the evaluation harness used to produce Tables II, III, and IV (note that Table IV requires API access to proprietary models and GPU resources for open-weight models). Production infrastructure, the agentic chat orchestration, and analyst-facing applications are not released, as they are part of an operational system in continuous use.

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